

Open Set Recognition for Machinery Fault Diagnosis

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Abstract—AI tasks based on deep neural networks have been widely applied in industrial applications, such as process control, quality inspection or predictive maintenance. Deep neural network classifiers are particularly successful, as they provide powerful and reliable algorithms for many applications such as object recognition and fault diagnosis. However, most deep classifier applications are not able to recognize class samples that are beyond the scope of their training data. Samples of unknown classes (denoted as open set data) lead to significant drops in performance, as the output of deep classifiers is limited to the known classes of the training data (denoted as closed set data). This paper presents a method to recognize open set samples without changing the neural network architecture, the training process, nor the trained models. In our method, we firstly train a neural network for normal closed set fault diagnosis. Then we compare the feature maps of testing samples and known class samples during inference using local outlier factor to recognize open set samples. We evaluate our method with two public datasets and show that our method can increase the overall accuracy by 40% when classifying open set data. Besides, we also compared our method to the state-of-the-art open set recognition approach for fault diagnosis applications and the results show that our method leads to better F1-scores.

Index Terms—fault diagnosis, open set recognition, deep learning

I. INTRODUCTION

Deep learning has been showing great performance for machine learning tasks on highly complex data due the superior feature learning capabilities of deep neural networks. Deep learning techniques have been widely applied to industrial applications nowadays, such as quality assurance, equipment maintenance and yield enhancement [1]–[3]. These techniques are well-suited for the industrial fault diagnosis as well, which is normally based on supervised classifiers since exact fault types should be predicted by the models rather than simply tell abnormal samples apart from the normal ones in fault detection. Similar to most classification tasks accomplished by deep learning techniques, the models are usually trained using pre-collected datasets with static fault types, which is referred to as a *closed set*. However, it is often hard or almost impossible to collect extensive distinctive fault samples for training in practice, as some fault patterns can only be known after they appear on real machinery. These unknown fault patterns are referred to as *open set*. In this case, the deep neural network classifiers fail, as they are not able to reject

unknown class samples, but output predictions that belong to the existing, trained classes. These situations can lead to severe consequences since people can be misguided and take wrong maintenance actions.

In this paper, we present a method that enables deep classifiers to reject open set samples for fault diagnosis applications. The underlying problem is known as *open set recognition* (OSR). As basis, we train convolutional neural networks (CNNs) for fault diagnosis using two public datasets describing the working status of a gearbox and a motor bearing respectively. Our method then utilizes intermediate feature maps of the CNN classifiers to estimate the distance of samples during inference to the known classes. Based on an improved distance metric, our machinery fault diagnosis application can then recognize, reject, and notify outliers.

To the best of our knowledge, this is one of the first works that explores the open set recognition problem for fault diagnosis with deep learning techniques [4]. The main contribution of this paper is the proposal of a novel open set recognition method for machinery fault diagnosis in the context of deep learning and the evaluation of the method on two public datasets.

II. RELATED WORK

In this section, we first review the existing deep learning techniques applied to machinery fault diagnosis problems. Furthermore, we briefly introduce the approaches on how to deal with the *unknowns* in deep learning, in which OSR a subset is. Finally, we summarize the approaches for open set recognition in the context of deep learning as well as the related application in machinery fault diagnosis.

A. Deep Learning for Machinery Fault Diagnosis

Fault diagnosis with machine learning techniques aims to output the failure types through analysing the sensory measurements. In the context of deep learning, Tamilselvan et. al. firstly proposed to classify machinery faults using deep belief networks in [2]. Afterwards, diverse neural network architectures have been continuously utilized for these problems. Janssens et. al. proposed the use of convolutional neural networks to bearing fault diagnosis in [5]. Yang et. al. applied long-short-term memory for wind turbine failure

classification in [6]. And Liu et. al. utilized gated recurrent unit for rolling bearings fault diagnosis [7]. Furthermore, additional improvements to standard deep learning techniques have been invited to fault diagnosis in order to mitigate issues seen in practice, such as deep transfer learning for small-volume data and dynamic working conditions [8], semi-supervised learning for insufficient labeled data [9] as well as encrypted deep learning for data secrecy [10]. However, among all these, open set recognition has yet rarely been discussed.

B. Unknowns in Deep Learning

The concept of *unknowns* here refers to not only the open set samples, but also all the samples that derived from different statistical distributions from the training data (i.e., out of distribution). Consequently, open set samples are a subset of such *unknowns* and specific for classification tasks. *Uncertainty quantification* (UQ), *out of distribution detection* (ODD), and *open set recognition* (OSR) are the three most common approaches for unknown detection problems.

UQ aims at enabling the models to output not only the predicted results, but also their confidences on the predictions. It is assumed that the confidences can be low when the models receive unknown samples as input. The well-known method categories for UQ are Bayesian neural networks [11], deep ensembles [12], and direct uncertainty learning [13]. Peng et al. applied Bayesian neural network based uncertainty estimation methods for machinery health prognostics in [14]. Li et al. proposed a Bayesian neural network framework to characterize the uncertainties in remaining useful life prediction tasks in [15]. However, uncertainties can originate from multiple sources, i.e., aleatory uncertainty (data noise) and epistemic uncertainty (improper model selection as well as unknown testing data). Therefore, uncertainty quantification cannot present explicit results for OSR problems because it cannot decide whether it is open set or not.

As the name implies, ODD is to detect samples that are generated by different distributions than the one of the training samples. There are three approach categories in literature. Hendrycks et al. have firstly proposed the baseline method for ODD in image recognition tasks through inspection of the *softmax* outputs, which are more averaged for out-of-distribution samples [16]. However, this method has low accuracy, since the neural networks can assign high prediction probabilities to out-of-distribution samples when they are uncalibrated. Liang et al. proposed their *ODIN* method to calibrate deep neural networks using temperature scaling before outputting the *softmax* probabilities. Besides detecting out-of-distribution samples using the predicted class probabilities, Lee et al. proposed an ODD method using the distances between in distribution and out of distribution feature maps [17]. If test sample features are far from the training data features, then it is out of distribution. ODD can also be realized through auxiliary models. Generative or unsupervised models, such as generative-adversarial networks, are often applied specifically for ODD, which is similar as for anomaly detection [18].

C. Open Set Recognition

The focus of this paper is OSR, which is specifically for classification tasks. It is defined as telling new class samples apart from known ones. In classification problems, ODD and OSR can be alternatives to each other. In the following text, we denote the known classes of the original classifier training as *inliers* and the unknown classes of the open set as *outliers*.

The most well-known method for OSR is *OpenMax* proposed by Bendale et al. in [19]. In OpenMax, each closed set class is represented using the mean activation vectors (MAVs), which is the mean of the activation vectors (i.e., the layer output) in the last fully connected layer of neural networks. Further, the distances of the activation vectors of the training samples and their corresponding MAVs are computed and the largest distances are modeled using a *Weibull distribution*, which is usually utilized to model rare events. During testing, the test sample is firstly predicted using the original classifier. The distance between the activation vector of testing sample and MAV of the predicted class is then used to inference the *Weibull* distribution. The higher the cumulative distribution function is, the more likely the testing sample is outlier. Later, Hassen et al. improved *OpenMax* through enclosing the data features from the same classes by adjusting the loss functions during training to increase the OSR accuracy [20]. All the above are based on discriminative methods and OSR can be realized through generative methods as well. Generative models can be utilized to mimic the open set data that was used to train the classifiers in order to enhance their abilities to recognize the outliers. Neal et. al. proposed to train the classifiers with one additional *outlier* class using the samples generated using generative adversarial networks [21]. Furthermore, Oza et. al. proposed their CA2E method in which a conditional auto-encoders are trained using inliers and the

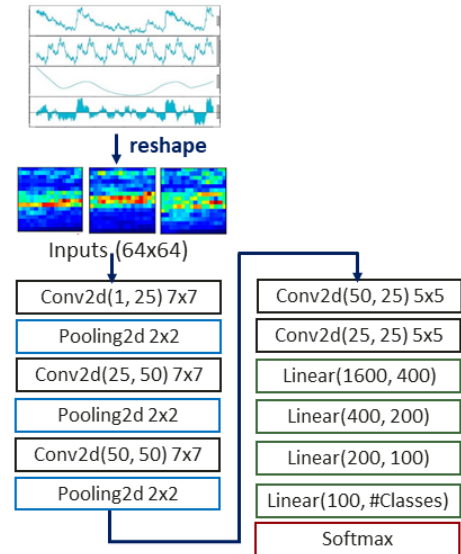


Fig. 1: The time series data is reshaped into 2D images and fed into CNN classifiers with the shown layer-wise configuration.

outliers can be detected through the reconstruction errors [22].

OSR and ODD have started to draw attention in the fields of autonomous driving and medical applications [23]–[25]. However, the only one work on OSR for fault diagnosis and even industry cases is [4], which is completely based on *OpenMax*.

III. METHODS

We first present the basic deep learning approach for normal machinery fault diagnosis. We then introduce our open set recognition method for unknown samples based on the neural networks developed.

A. CNN Classifiers for Fault Diagnosis

Convolutional neural networks show outstanding performance in feature learning and machinery fault diagnosis. Thus, we adopt the CNN architecture for our classifiers and as the basis for the open set recognition methods in this paper. The network is composed of six convolutional layers and three fully connected layers. The inputs are time series windows that are reshaped to 64×64 images for the 2D convolutional layers. The pre-processing and layer-wise configuration is shown in Fig. 1. After reshaping and successive pooling layers, the network are expected to learn the connections between neighboring as well as distant data samples. The classifiers are trained on inlier samples only (see Section IV-A for the concrete datasets).

B. OSR Method with Mahalanobis distances

The OSR method with *Mahalanobis* distances was firstly proposed in [17] that relies on comparing the distances between the features of (open set) samples and the known class clusters. This method doesn't change the original structure of the neural networks nor require retraining.

In our settings, we use the feature maps from the outputs of the penultimate fully connected layer, i.e., Linear(200, 100) in Fig. 1, to represent the raw data samples and calculate the distances. It is defined in (1), where $\mathbf{x}$ represents a sample and μ and $\mathbf{S}$ are the mean and co-variance of a cluster, respectively. Compared to the *Euclidean* distance, the Mahalanobis distance also considers the shapes of clusters and yields better results for practical data [17].

$$M(\mathbf{x}) = \sqrt{(\mathbf{x} - \mu)^T \mathbf{S}^{-1} (\mathbf{x} - \mu)} \quad (1)$$

Our procedure for computing the distances is to first compute and store the cluster centers (i.e., μ_{cl} , $\mathbf{S}_{cl}$) for each training data class (i.e., $\mathbf{X}_c$) using the feature maps of the penultimate fully connected layer (i.e., $f_l(x_c)$ with $l = L - 2$) according to (2) and (3). Then for each input sample x , the Mahalanobis distances M_{cl} between their features $f_l(\mathbf{x})$ and each cluster center are computed as in (4).

$$\mu_{cl} = \frac{1}{|\mathbf{X}_c|} \sum_{\mathbf{x}_c \in \mathbf{X}_c} \mathbf{f}_l(\mathbf{x}_c) \quad (2)$$

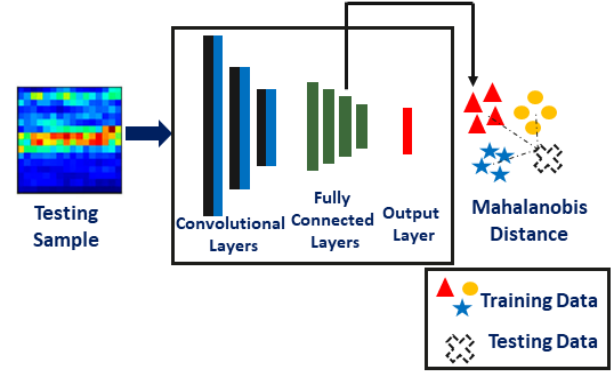


Fig. 2: Pipeline of open set recognition using neural network feature maps: The feature maps of the testing data are firstly read from the penultimate fully connected layer and then compared with the existing features maps from the training data of different classes. The testing samples that are far from any class clusters are seen as open set data.

$$\mathbf{S}_{cl} = \frac{1}{|\mathbf{X}_c|} \sum_{\mathbf{x}_c \in \mathbf{X}_c} (\mathbf{f}_l(\mathbf{x}_c) - \mu_{cl})^T (\mathbf{f}_l(\mathbf{x}_c) - \mu_{cl}) \quad (3)$$

$$M_{cl} = \sqrt{(\mathbf{f}_l(\mathbf{x}) - \mu_{cl})^T \mathbf{S}_{cl}^{-1} (\mathbf{f}_l(\mathbf{x}) - \mu_{cl})} \quad (4)$$

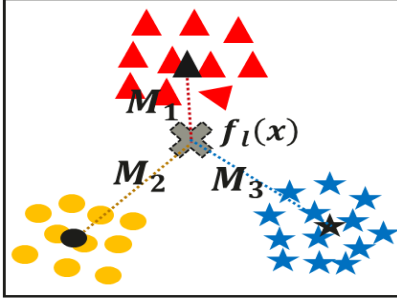
An input sample could now be considered an open set sample, if the minimum distance among all M_{cl} indicating the class prediction is larger than a predefined threshold T_{simple} (T_{simple} is the largest distance among the 95% closest samples to the class centers in our experiments). However, the sparsity of the class clusters is heterogeneous. For illustration, Table I shows the average distances of the training samples to their cluster centers of the *Paderborn* dataset (see IV-A) and it is easy to find that the average distances strongly differ among the classes. Therefore, solely thresholding the distances cannot offer promising accuracy. For example, a testing sample that is close to Class 10 may be predicted as outlier if the threshold is set under 2000 when considering the average distances of other classes.

C. Proposed OSR Method

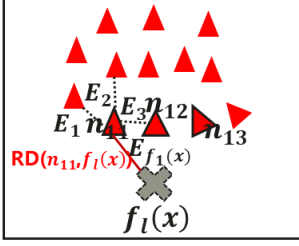
To tackle the issue with simple thresholding, we propose a new approach in which we combine the idea of the local outlier factor (LOF) [26] with the OSR method in III-B. LOF

Class	0	1	2	3	4
Avg. Dist.	98.26	4114.67	279.54	398.14	1672.57
Class	5	6	7	8	9
Avg. Dist.	398.21	1185.27	313.68	1818.37	142.69
Class	10	11	12	13	14
Avg. Dist.	3297.79	113.84	101.73	112.48	294.78

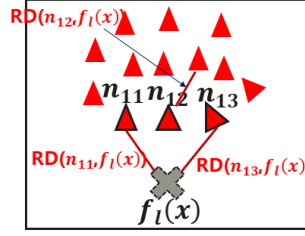
TABLE I: Intra-class average distances of the training data of the Paderborn dataset.



(a) The input sample features $f_l(\mathbf{x})$ are assigned to class 1 (red triangles) according to the smallest Mahalanobis distance among M_1 , M_2 and M_3 (black markers are class centers).



(b) The reachability distance between $f_l(\mathbf{x})$ and n_1 is the maximum Euclidean distances between n_{11} and its k neighbors as well as $f_l(\mathbf{x})$ ($k = 3$ in the figure for illustration).



(c) RD for all n_{1k} is computed in the same way. And the local reachability density of $f_l(\mathbf{x})$ is the reciprocal of the average of k neighbor RDs ($k = 3$ in the figure for illustration).

Fig. 3: Illustration on LOF computing for $f_l(\mathbf{x})$.

can better describe how far a sample from a given cluster is. We identify the closest class cluster c^* for the input sample $\mathbf{x}$ on the feature map (i.e., $f_l(\mathbf{x})$) through its Mahalanobis distances to the class centers as illustrated in Fig. 3a. $c^* = 1$ in the figure. Then the k nearest neighbors (i.e., $\mathbf{n}_{c^*1}, \mathbf{n}_{c^*2}, \dots, \mathbf{n}_{c^*k}$) of x belonging to the class cluster c^* are searched using the Euclidean distances. The reachability distance (RD) of a neighbor-sample pair is the maximum Euclidean distance between the neighbor $\mathbf{n}_i$ and its k nearest neighbors and $f_l(\mathbf{x})$. RD is shown in Fig. 3b and (5), where E_k represents the corresponding Euclidean distance metric.

The local reachability density (LRD) is the reciprocal of the average of k neighbor RDs as shown in (6), where the reference r can either be the sample features $f_l(\mathbf{x})$ or one of its k neighbors $\mathbf{n}_{c^*i}$. The farther the reference point is from its k neighbors, the smaller is the LRD value. LOF is finally defined as the fraction between the average LRD of the k neighbors $\mathbf{n}_i$ and the LRD of $f_l(\mathbf{x})$.

$$RD = \max(E_1, E_2, \dots, E_k, E_{f_l(\mathbf{x})}) \quad (5)$$

$$LRD_k = \left(\frac{1}{k} \sum_{i=1}^k RD(\mathbf{r}) \right)^{-1} \quad (6)$$

$$LOF_{c^*k}(f_l(\mathbf{x})) = \frac{\frac{1}{k} \sum_{i=1}^k LRD_k(\mathbf{n}_{c^*i})}{LRD_k(f_l(\mathbf{x}))} \quad (7)$$

The full method for OSR with *local outlier factor* is given in Algorithm 1. For each input sample $\mathbf{x}$, its closest class c^* is firstly predicted with the Mahalanobis distances between its feature map and each class cluster center. Then the LOF_{c^*k} with a chosen value k is computed and compared to a predefined threshold T_{LOF} (T_{LOF} is chosen to be 2 in our experiments). In theory, the greater the LOF value is, the more likely is the sample an outlier.

Algorithm 1 Open Set Recognition

Train classifier:

Training data $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_C\}$ for C classes
Convolutional neural network with L layers

Initialize:

Extract class feature maps $\mathbf{f}_l(\mathbf{X}_c)$ from layer $l = L - 2$
for each class $c \in \{1, 2, \dots, C\}$ **do**
 $\mu_{cl} = \frac{1}{|\mathbf{X}_c|} \sum_{\mathbf{x}_c \in \mathbf{X}_c} \mathbf{f}_l(\mathbf{x}_c)$
 $\mathbf{S}_{cl} = \frac{1}{|\mathbf{X}_c|} \sum_{\mathbf{x}_c \in \mathbf{X}_c} (\mathbf{f}_l(\mathbf{x}_c) - \mu_{cl})^T (\mathbf{f}_l(\mathbf{x}_c) - \mu_{cl})$
end for

for each input sample $\mathbf{x}$ do

 Read feature map $f_l(\mathbf{x})$ of layer $l = L - 2$
 for each class c with center $(\mu_{cl}, \mathbf{S}_{cl})$ **do**
 Calculate Mahalanobis distance $M_{cl}(f_l(\mathbf{x}))$
 end for
 Search for the closest class $c^* = \arg \min_c M_{cl}$
 Calculate $LOF_{c^*k} = LOF_{c^*k}(f_l(\mathbf{x}))^c$
 if $LOF_{c^*k} < T_{LOF}$ **then**
 $\mathbf{x} \leftarrow \text{inlier}$
 else
 $\mathbf{x} \leftarrow \text{outlier}$
 end if
end for

IV. EVALUATION

A. Datasets

We use two public machinery fault diagnosis datasets to evaluate the methods discussed in III, i.e., the *PHM Society Competition* (PHM09) dataset [27] and the *Paderborn University* dataset [28]. Both datasets are time series sensory measurements. Overall, there are 5, 120, 020 samples and 199, 240 samples for each fault class in the datasets, respectively.

The PHM09 dataset describes the gear fault modes of a running gearbox using vibration measurements. The gearbox consists of six bearings, three shafts, and four gears (two spur gears and two helical gears). There are six fault modes for each spur gear and eight fault modes for each helical gear, i.e., 28 fault classes in total. Furthermore, the data was collected under five running speeds (30Hz, 35Hz, 40Hz, 45Hz, 50Hz) and two loading modes (high, low), i.e., ten working conditions in total. We conduct experiments on the ten working conditions independently and all the 28 fault classes are taken.

The Paderborn dataset are motor current measurements used to diagnose the rolling bearings. There are four types of working conditions (N15M07F10, N09M07F10, N15M01F10, N15M01F04) corresponding to different rotational speed (N), load torque (M), and radial force (F). In the following text, they are denoted as conditions 1–4. There are 26 fault classes in total with different damage types for inner ring, outer ring, and both rings.

We randomly split the fault classes in each dataset to constitute inlier and outlier samples. Half of the classes are assumed to be inliers and the other half are outliers (i.e., 14 inlier classes and 14 outlier classes for the PHM09 dataset and 13/13 for the Paderborn dataset).

B. CNN Classifiers

Two CNN classifiers are trained using inlier samples only from each dataset. 60% of the inlier samples are used for training, 20% for validation, and the remaining 20% for testing.

For the open set experiments with inliers and outliers, we take 20% of the samples from the entire dataset (i.e., 20% inlier samples for normal CNN testing as talked above altogether with 20% randomly selected outlier samples). In such case, the volumes of inlier and outlier samples used for evaluation are roughly the same. And it is in principle more strict situation than in practice since outliers are relatively rare events. However, such experimental configuration can show that our method can be well scalable.

We use accuracy, precision, and recall as metrics for evaluation. The accuracy here is the ratio between the number of correctly classified samples to the number of the total samples as shown in (8). N_{correct} is the number of correctly classified samples and N is the number of all samples. Since the models are multi-class classifiers, the precision and recall are the average of accuracy_c , precision_c , and recall_c of each class c as shown in (9) and (10). C is the number of classes.

$$\text{accuracy} = \frac{N_{\text{correct}}}{N} \quad (8)$$

$$\text{recall} = \frac{1}{C} \sum_{c=1}^C \text{recall}_c \quad (9)$$

$$\text{precision} = \frac{1}{C} \sum_{c=1}^C \text{precision}_c \quad (10)$$

For the classifiers with OSR, it is assumed to be a $C + 1$ classifier, where a collective class for all outliers is added. For the original models without OSR, it is easy to understand that none of the outliers can be correctly classified. The accuracy, precision, and recall for this new class are therefore zero.

The results for each dataset are shown in Table II and Table III, respectively. It can be seen that outliers clearly degrade the model performance through all the three metrics, especially for accuracy and precision. Since outliers are all wrongly assigned inlier class labels, the false positive value increases and precision drops in each class. Accuracy drops with the increase of the outlier samples, as they cannot be

recognized without OSR. The last three rows of Table II and Table III show that our OSR method in Alg. 1 can significantly improve performance (e.g., by 40% for accuracy) again, as outliers can now be recognized – even without any modification to the classifier networks.

C. LOF Improvement

We also evaluate how much the integration of LOF improves performance over simple Mahalanobis distance thresholding. We take precision and recall as metrics since OSR can be intuitively regarded as binary classification problems. And F1-score indicates the balance between accuracy and recall as shown in (11). Furthermore, we have re-implemented the OSR approach in [4] and tested on the above two datasets. The evaluation results are shown in Table IV and Table V.

$$F1\text{-score} = \frac{2}{\text{precision}^{-1} + \text{recall}^{-1}} \quad (11)$$

The results show that the OSR method with LOF proposed in this paper can show better performance on both experimental datasets than simple thresholding. According to Table V and Table IV, precision increases with LOF. It is because less outliers that are located at the class cluster borders can be recognized and the false positive value decreases (inliers are assumed to be positive in the experiments). Recall drops slightly since some inliers that are relatively "isolated" from the clusters are wrongly recognized as outliers and the false negative value increases. Overall, however, LOF increases the F1-score, which is the harmonic mean of precision and recall and reveals the recognition accuracy. Moreover, the OSR method in [4] is actually equivalent to thresholding the *Euclidean* distances between the testing samples and the class cluster centers and it ignores the geometry of the inlier class clusters compared with Mahalanobis distance [17]. Its precision and F1-scores are therefore even lower than simple thresholding.

V. CONCLUSION

This paper aims at introducing open set recognition (OSR) techniques to industrial applications. While CNN classifiers perform splendidly on complex machinery and sensory data, they are usually limited to the known classes of the training data. When outliers appear, for instance because of different usage patterns, machine fatigue, or simply so far unobserved conditions, their performance drops.

We propose an OSR method that significantly improves CNN classifier performance on open set data, while it does not require changes to existing deep learning models. Our evaluation shows that a combination of Mahalanobis distance metrics and the local outlier factor (LOF) performs well for machinery fault diagnosis tasks. The positive effect of LOF indicates that outliers can be very similar to known classes, which is a key challenge for OSR.

So far, our OSR method is only tested on CNNs, but has the potential to be generalized to other types of deep neural networks. Moreover, additional OSR or ODD strategies can be

Results \ Conditions	30Hz High	30Hz Low	35Hz High	35 Hz Low	40Hz High	40Hz Low	45Hz High	45 Hz Low	50Hz High	50Hz Low
Precision (closed)	99.94%	99.98%	99.98%	99.96%	99.95%	99.97%	99.96%	100%	99.98%	99.99%
Recall (closed)	99.94%	99.98%	99.98%	99.96%	99.95%	99.97%	99.96%	100%	99.98%	99.99%
Accuracy (closed)	99.94%	99.98%	99.98%	99.96%	99.95%	99.97%	99.96%	100%	99.98%	99.99%
Precision (open)	57.80%	62.56%	69.25%	58.15%	58.63%	60.17%	57.42%	64.11%	61.53%	57.66%
Recall (open)	93.28%	93.31%	93.32%	93.29%	93.29%	93.31%	93.30%	93.33%	93.31%	93.24%
Accuracy (open)	49.84%	49.91%	50.03%	49.86%	50.30%	49.93%	49.92%	49.58%	49.90%	49.82%
Precision (with Alg. 1)	90.52%	86.50%	93.85%	91.47%	87.10%	88.30%	94.18%	93.51%	89.34%	86.04%
Recall (with Alg. 1)	94.52%	89.28%	95.43%	94.78%	94.65%	92.13%	96.26%	95.92%	94.57%	90.72%
Accuracy (with Alg. 1)	91.04%	91.30%	92.62%	92.30%	88.30%	88.01%	93.90%	93.83%	89.48%	85.92%

TABLE II: Classification accuracy of inlier samples tested using CNN and PHM09 dataset. **Upper part:** classification performance of the CNNs for inlier samples only. **Middle part:** classification performance of the CNNs tested using inlier and outlier samples. **Bottom part:** classification performance of the CNNs with OSR tested using inlier and outlier samples.

Results \ Conditions	1	2	3	4
Precision (closed)	99.78%	99.34%	99.93%	99.95%
Recall (closed)	99.79%	99.34%	99.92%	99.95%
Accuracy (closed)	99.82%	99.34%	99.92%	99.96%
Precision (open)	60.55%	59.98%	53.10%	57.41%
Recall (open)	92.69%	92.80%	92.78%	92.82%
Accuracy (open)	49.94%	49.94%	49.96%	50.96%
Precision (with Alg. 1)	81.87%	85.50%	86.53%	85.74%
Recall (with Alg. 1)	90.67%	92.20%	93.10%	90.30%
Accuracy (with Alg. 1)	80.73%	83.85%	86.78%	85.06%

TABLE III: Classification accuracy of inlier samples tested using CNN and Paderborn dataset. **Upper part:** classification performance of the CNNs for inlier samples only. **Middle part:** classification performance of the CNNs tested using inlier and outlier samples. **Bottom part:** classification performance of the CNNs with OSR tested using inlier and outlier samples.

introduced, such as input permutation [17] to further improve results. Last but not least, it should become possible to automatically react to outlier recognition with corresponding model updates, for instance through continual learning or incremental learning techniques.

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Conditions Results	30Hz High	30Hz Low	35Hz High	35Hz Low	40Hz High	40Hz Low	45Hz High	45Hz Low	50Hz High	50Hz Low
Precision ([17])	73.30%	74.75%	79.07%	66.07%	75.89%	70.24%	86.02%	82.74%	70.84%	70.80%
Recall ([17])	99.70%	98.50%	99.81%	99.76%	99.34%	99.10%	99.81%	99.91%	99.44%	83.95%
F1-Score ([17])	83.82%	85.04%	88.24%	79.50%	86.04%	82.21%	92.40%	90.52%	90.52%	76.82%
Precision (Alg. 1)	90.93%	82.96%	93.41%	90.48%	87.67%	85.10%	91.80%	92.14%	92.88%	82.65%
Recall (Alg. 1)	90.25%	92.10%	90.91%	95.04%	82.29%	93.03%	96.35%	95.73%	83.33%	91.41%
F1-Score (Alg. 1)	90.59%	87.29%	92.14%	92.70%	93.08%	88.89%	94.02%	92.90%	88.63%	86.81%
Precision ([4])	54.02%	52.51%	58.79%	60.59%	53.64%	52.82%	55.22%	61.33%	53.89%	56.42%
Recall ([4])	99.93%	99.91%	99.96%	99.93%	99.97%	99.90%	99.96%	99.94%	99.98%	99.87%
F1-Score ([4])	70.13%	68.84%	74.04%	75.44%	69.82%	69.10%	71.14%	76.01%	70.03%	72.11%

TABLE IV: Evaluation results for OSR using PHM09 dataset. **Upper part:** OSR with simple threshold. **Middle part:** OSR with LOF. **Bottom part:** OSR with method in [4]

Conditions Results	1	2	3	4
Precision ([17])	67.43%	70.83%	73.52%	64.93%
Recall ([17])	98.49%	99.37%	98.77%	99.42%
F1-Score ([17])	80.06%	82.71%	84.30%	78.55%
Precision (Alg. 1)	78.16%	79.80%	84.04%	81.69%
Recall (Alg. 1)	86.71%	88.91%	89.29%	89.55%
F1-Score (Alg. 1)	82.21%	84.11%	86.59%	85.44%
Precision ([4])	54.63%	56.04%	57.54%	56.77%
Recall ([4])	99.98%	99.95%	99.95%	99.92%
F1-Score ([4])	70.66%	71.81%	73.04%	72.41%

TABLE V: Evaluation results for OSR using Paderborn dataset. **Upper part:** OSR with simple threshold. **Middle part:** OSR with LOF. **Bottom part:** OSR with method in [4].

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